Introduction to Probability and Statistics
(EE 354 / CE 361)

Discrete RVs: Probability Mass Functions, Common Discrete RVs, and Expected Value

Today's Agenda

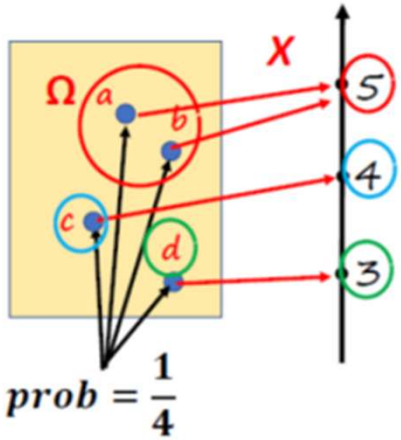
- Probability Mass Functions
- Common Discrete RVs
  - Bernoulli Random Variable
  - Binomial Random Variable
  - Geometric Random Variable
  - Discrete Uniform Random Variable
- (time permitting) Expected Value

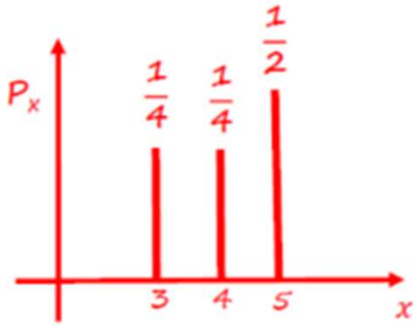
Probability Mass Function (PMF) of a Discrete Random Variable

- How to characterize/describe a random variable
  - The most important way to characterize a random variable is through the probabilities of the values it can take.
  - For a discrete random variable X, this information is captured by its **Probability Mass Function (PMF)**, denoted by: p_X

In particular, if x is any possible value of X, the **probability mass** of x, denoted $p_X(x)$, is the probability of the event $\{X = x\}$ consisting of all outcomes that give rise to a value of X equal to x:

$$p_X(x) = \mathbf{P}(\{X = x\}) = \mathbf{P}(\{w \in \Omega \text{ s.t. } X(w) = x\})$$



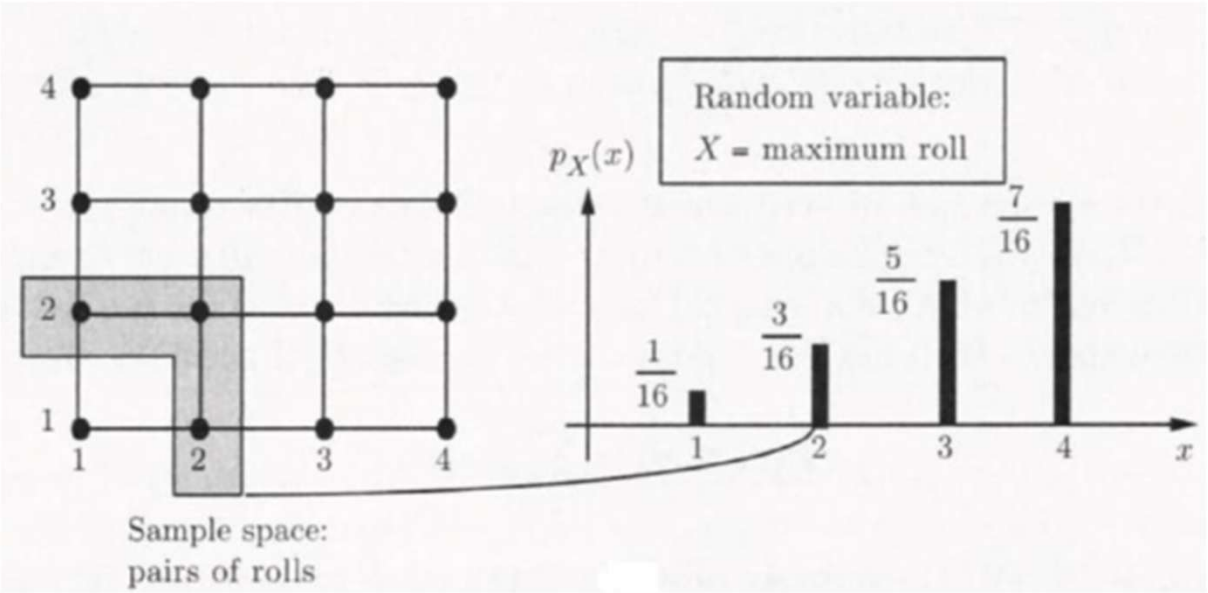


Calculation of the PMF of a Random Variable

For each possible value x of X:

1. Collect all the possible outcomes that give rise to the event $\{X = x\}$.
2. Add their probabilities to obtain $p_X(x)$.

- Example
  - Two rolls of a fair tetrahedral die
  - Random Variable: X = Maximum of two rolls

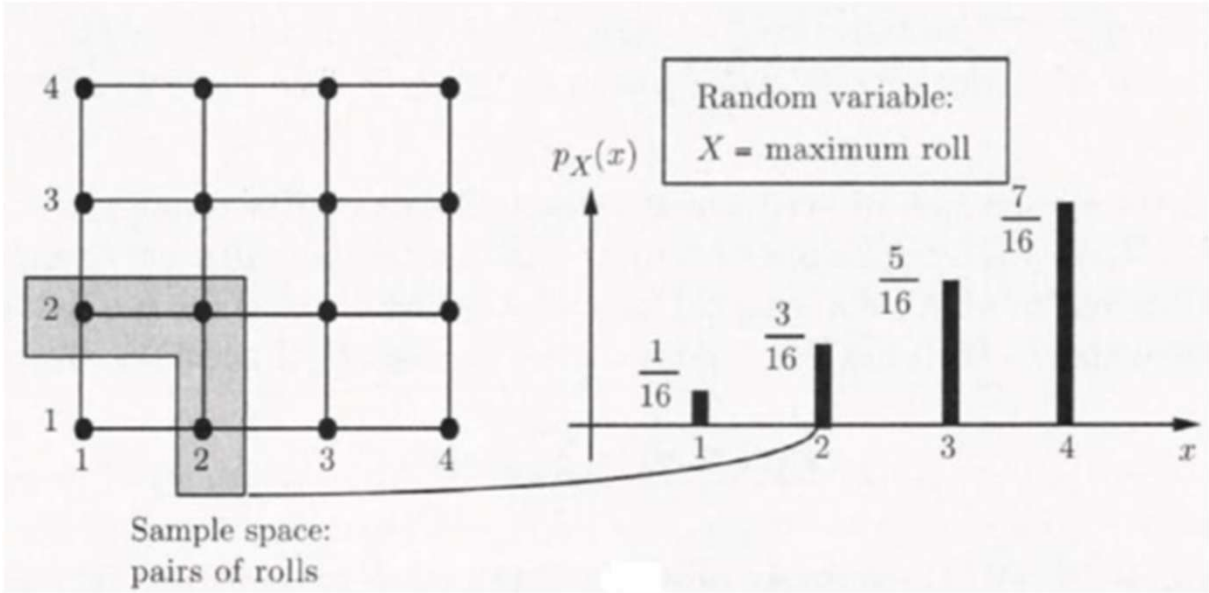


Properties of the PMF of a Random Variable

- Properties:

$$p_X(x) \geq 0 \qquad \sum_x p_X(x) = 1$$

- Example
  - Two rolls of a fair tetrahedral die
  - Random Variable: X = Maximum of two rolls



Axioms:

a) $\mathbf{P}(A) \geq 0$

b) $\mathbf{P}(\Omega) = 1$

For disjoint events

c) $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B)$

Properties of the PMF of a Random Variable

for any set S of possible values of X

$$\mathbf{P}(X \in S) = \sum_{x \in S} p_X(x).$$

- Example
  - Two independent tosses of a fair coin
  - Random Variable: X = Number of Heads in two tosses
  - Probability of at least 1 Head = ?

$$\mathbf{P}(X > 0) = \sum_{x=1}^{2} p_X(x) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}.$$

Example: Functions of Random Variables

Example 2.1. Let $Y = |X|$ and let us apply the preceding formula for the PMF p_Y to the case where

$$p_X(x) = \begin{cases} 1/9. & \text{if } x \text{ is an integer in the range } [-4, 4], \\ 0. & \text{otherwise:} \end{cases}$$

What is $p_Y(y)$?

Example: Functions of Random Variables

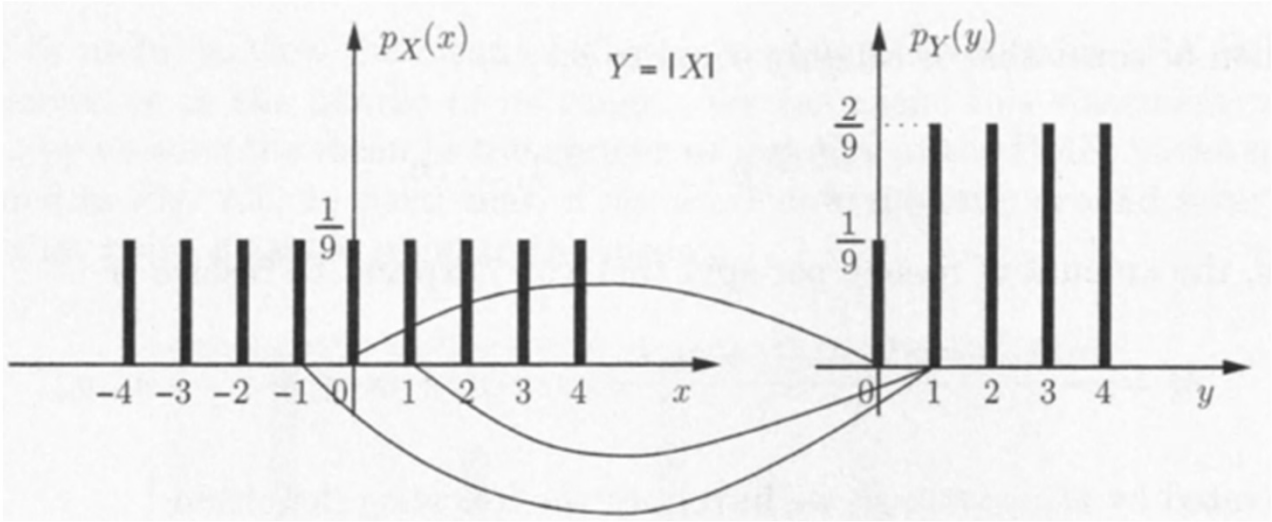
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What is $p_Y(y)$?

PMF of Y is

$$p_Y(y) = \begin{cases} 2/9. & \text{if } y = 1.2, 3, 4, \\ 1/9. & \text{if } y = 0. \\ 0. & \text{otherwise.} \end{cases}$$



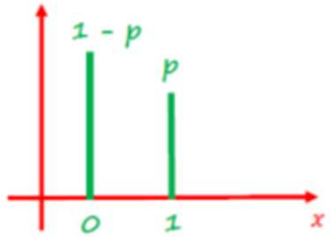
Bernoulli Random Variable

- Experiment
  - The toss of a coin which comes up heads with probability p and tails with probability $(1 - p)$.
- Sample Space
  - {Head, Tail}

$$X = \begin{cases} 1. & \text{if a head,} \\ 0. & \text{if a tail.} \end{cases}$$

- Random Variable
  - The RV takes value 1 if the outcome is head and takes the value 0 if the outcome is a tail.
- PMF

$$p_X(k) = \begin{cases} p, & \text{if } k = 1, \\ 1 - p, & \text{if } k = 0. \end{cases}$$



- Model of:
  - Used to model a generic probabilistic situation with just two possible outcomes
  - Examples

The state of a telephone at a given time that can be either free or busy.

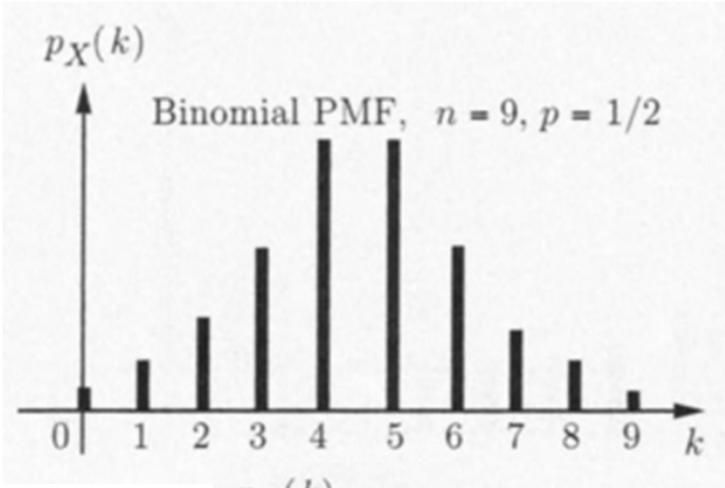
A person who can be either healthy or sick with a certain disease.

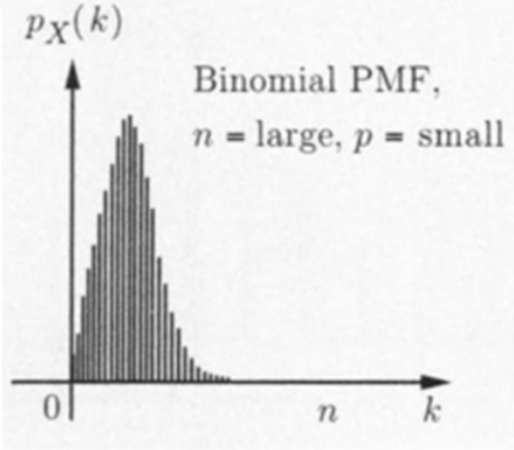
The preference of a person who can be either for or against a certain political candidate.

- Parameters: $\boldsymbol{p \in [0, 1]}$

Binomial Random Variable

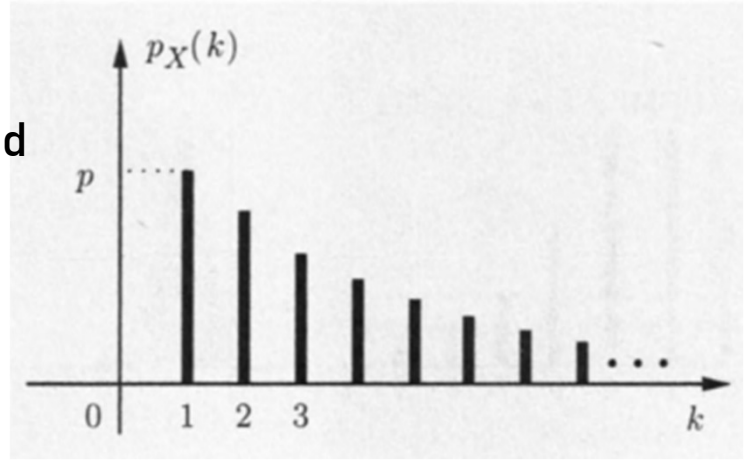
- Experiment
  - n independent tosses of a coin where at each toss, the coin comes up a head with probability p and a tail with probability $(1 - p)$.
- Sample Space
  - Set of Head (H) and Tail (T) sequences of length n
- Random Variable
  - The number of Heads in the n-toss sequence
- PMF
  - $\boldsymbol{p_X(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad , k = 0, 1, 2, \ldots, n}$
- Model of:
  - Number of successes in a given number of independent trials
- Parameters: $\boldsymbol{n \in \mathbb{Z}^{++}}$, $\boldsymbol{p \in [0, 1]}$





Geometric Random Variable

- Experiment
  - (infinitely) many independent tosses of a coin where at each toss, the coin comes up a head with probability p and a tail with probability $(1 - p)$.
- Sample Space
  - Set of infinite sequences of Head (H) and Tail (T)
- Random Variable
  - The number of tosses until the first head



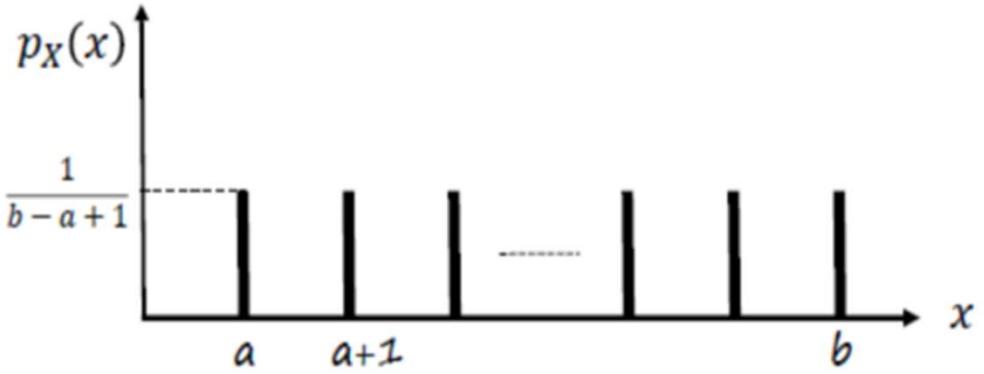
- PMF
  - $\boldsymbol{p_X(k) = (1-p)^{k-1}p \quad , k = 1, 2, \ldots}$
- Model of:
  - Number of independent trials until a success
  - Example: the number of times a student will take the test before passing the test
- Parameters: $\boldsymbol{p: 0 < p < 1}$

This is a legitimate PMF because

$$\sum_{k=1}^{\infty} p_X(k) = \sum_{k=1}^{\infty} (1-p)^{k-1} p = p \sum_{k=0}^{\infty} (1-p)^k = p \cdot \frac{1}{1-(1-p)} = 1.$$

Discrete Uniform Random Variable

- Experiment
  - Pick one out of a range of contiguous integer values $(a, a+1, \ldots, b)$ with equal probability.
- Sample Space
  - $\{a, a+1, \ldots, b\}$
- Random Variable
  - For each integer in the sample space
    $X(w) = w$



- PMF
$$p_X(k) = \begin{cases} \frac{1}{b-a+1}, & \text{if } k = a, a+1, \ldots, b. \\ 0. & \text{otherwise.} \end{cases}$$
- Model of:
  - Used to model a set of equally likely outcomes
- Parameters: $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{Z}$

Expectation/Expected Value/Mean of a Random Variable

- Motivation
  - The PMF of a random variable X provides us with several numbers, the probabilities of the possible values of X.
  - It is often desirable to summarize this information in a single representative number.
  - This is accomplished by the Expectation of X, which a weighted (in proportion to probabilities) average of the possible values of X.
- Definition

Expectation

We define the **expected value** (also called the **expectation** or the **mean**) of a random variable X, with PMF p_X, by

$$\mathbf{E}[X] = \sum_x x p_X(x).$$

- Example

Example 2.2. Consider two independent coin tosses, each with a 3/4 probability of a head, and let X be the number of heads obtained. This is a binomial random variable with parameters $n = 2$ and $p = 3/4$. Its PMF is

$$p_X(k) = \begin{cases} (1/4)^2, & \text{if } k = 0, \\ 2 \cdot (1/4) \cdot (3/4), & \text{if } k = 1, \\ (3/4)^2, & \text{if } k = 2, \end{cases}$$

Expectation/Expected Value/Mean of a Random Variable

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$$\mathbf{E}[X] = 0 \cdot \left(\frac{1}{4}\right)^2 + 1 \cdot \left(2 \cdot \frac{1}{4} \cdot \frac{3}{4}\right) + 2 \cdot \left(\frac{3}{4}\right)^2 = \frac{24}{16} = \frac{3}{2}.$$

It is useful to view the mean of X as a "representative" value of X, which lies somewhere in the middle of its range. We can make this statement more precise, by viewing the mean as the **center of gravity** of the PMF, in the sense explained in Fig. 2.7. In particular, if the PMF is symmetric around a certain point, that point must be equal to the mean.

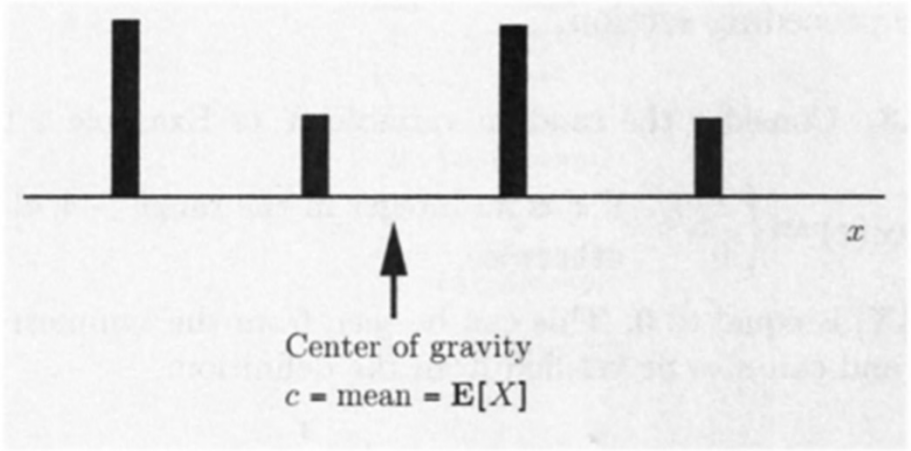


Figure 2.7: Interpretation of the mean as a center of gravity. Given a bar with a weight $p_X(x)$ placed at each point x with $p_X(x) > 0$, the center of gravity c is the point at which the sum of the torques from the weights to its left is equal to the sum of the torques from the weights to its right:

$$\sum_x (x - c)p_X(x) = 0.$$

Thus, $c = \sum_x xp_X(x)$, i.e., the center of gravity is equal to the mean $\mathbf{E}[X]$.